(a)

Configuration-level difference in mean outer-fold ROC-AUC

No-SMOTE higher

SMOTE higher

Source configuration

Post1 +0.024
Pre +0.020
T2 +0.004
ADC +0.003
Post2 -0.002

Single

Post2 + T2 +0.009
Post1 + T2 +0.007
Post1 + Pre +0.006
ADC + Post2 +0.004
Post1 + Post2 +0.002
ADC + Post1 -0.001
Post2 + Pre -0.002
ADC + Pre -0.006
Pre + T2 -0.007
ADC + T2 -0.007

Pairwise

Post1 + Post2 + Pre +0.025
Post2 + Pre + T2 +0.007
ADC + Pre + T2 +0.005
ADC + Post2 + Pre +0.003
ADC + Post1 + Post2 +0.002
ADC + Post2 + T2 -0.002
Post1 + Pre + T2 -0.006
Post1 + Post2 + T2 -0.006
ADC + Post1 + T2 -0.006
ADC + Post1 + Pre -0.013

Triple

ADC + Post2 + Pre + T2 +0.011
ADC + Post1 + Post2 + T2 +0.007
ADC + Post1 + Pre + T2 +0.004
Post1 + Post2 + Pre + T2 +0.004
ADC + Post1 + Post2 + Pre +0.003

Quadruple

Pre + Post1 + Post2 + ADC + T2 -0.005

All-five

−0.03 −0.02 −0.01 0.00 0.01 0.02 0.03

Δ Mean outer-fold ROC-AUC (SMOTE - no-SMOTE)

(b)

Wins by feature-fusion level

SMOTE higher

No-SMOTE higher

Number of configurations

0 1 2 3 4 5 6

Single: 4 | 1 — n=5
Pairwise: 5 | 5 — n=10
Triple: 5 | 5 — n=10
Quadruple: 5 | 0 — n=5
All-five: 0 | 1 — n=1